Scientific Research Methodology

8. Research misconduct: what is it and how can we avoid it?

Beniamin Bogosel

Faculty of Exact Sciences
Aurel Vlaicu University of Arad

Introduction

"Misconduct" = inappropriate or wrongful behaviour

Research misconduct = conducting research improperly

- fabrication or falsification of evidence
- plagiarism
- questionable research practices

Why discuss this?

- To recognise that science must be conducted with integrity;
- results must be reproducible;
- scientists must be trustworthy

Types of misconduct

1. Fabrication of evidence
   - inventing datasets to make it easier to derive particular results
   - Example: Diederik Stapel
     - Fabricated data for dozens of papers
     - More than 50 paper retractions
     - **Retraction =** withdrawal of a published paper shown to be false
2. Falsification of materials, data, processes, or results
   - Hwang Woo-suk: cloning using stem cells, 2004–2005
   - the author falsified data claiming successful cloning of human cells
   - his papers were retracted from the renowned journal *Science*

Types of misconduct

3. Plagiarism: using ideas from other work without citation
   - Jan Hendrik Schön: nanotechnology, advanced physics, superconductors
   - in 2001, he announced a molecular-scale transistor: a result with enormous implications that drew attention to his publications and led to the discovery of anomalies
   - data and results from some experiments were copied directly and presented as belonging to other experiments

Types of misconduct

4. Questionable research practices:
   - p-hacking
   - HARKing: Hypothesizing After Results are Known—formulating hypotheses after the results are known
   - selective reporting
   - inadequate data handling
   - failure to disclose conflicts of interest

What motivates fraudulent research behaviour?

- Pressure to publish: "publish or perish"
- Competition for funding
- prestige and career advancement
- Cognitive bias (confirmation bias, observer bias)
- superficial supervision (of doctoral students, for example); lack of appropriate skills and training in the field
- studies are not conducted reproducibly

Famous cases: Andrew Wakefield

- a fraudulent study linking a vaccine to autism
- data manipulation
- undisclosed financial conflicts of interest
- consequences: vaccination rates declined; a public health crisis; the paper was retracted
- the author lost his medical licence

Famous cases

Paolo Macchiarini: surgeon

- implanted synthetic tracheas without sufficient evidence or ethical approval
- patient deaths

Brian Wansink: research on marketing and consumption

- "food psychology"
- p-hacking, HARKing, data manipulation
- numerous paper retractions

Famous cases: Francesca Gino

- former Harvard Business School professor who studied "honesty"
- data falsification
- Example of a finding: signing at the beginning reduces the risk of dishonest behaviour
- Example of a finding: "Evil Genius? How Dishonesty Can Lead to Greater Creativity"
- Co-authors began re-examining more than 140 collaborations: in over 60%, co-authors had no access to the data
- Plagiarism allegations: some of her books contained passages from other sources (bachelor's or master's theses not supervised by Gino), papers, chapters, and magazine articles

Misconduct in practice

- poor study design $\longrightarrow$ fabricated, manipulated, or selectively chosen data
- data collection $\longrightarrow$ inaccurate reporting, cherry-picking
- statistical analysis $\longrightarrow$ p-hacking, selective reporting
- publication $\longrightarrow$ plagiarism, figure manipulation
- peer review $\longrightarrow$ conflicts of interest

Warning signs

- data that are too perfect (statistical anomalies, a perfect match with mathematical laws)
- raw data are unavailable
- negative results are hidden
- pressure from supervisors to obtain the expected results

Prevention: personal practices

- preregistering studies
- keeping detailed records of laboratory research
- version control for code and data
- independent replication (within the laboratory and externally)
- sound ethical principles for scientific publishing

Prevention: institutional practices

- research integrity training
- formal data-sharing requirements
- ethics committee review for clinical studies and animal research

Prevention: statistical and technical tools

- software for detecting plagiarism and image manipulation
- open repositories for depositing data

Activity

Hypothetical example

A doctoral student collects 200 data points; analysis does not reveal the desired effect. The supervisor suggests running more tests on subgroups until a "significant" result appears.

- Is this data fabrication, data falsification, or a questionable research practice?
- How should the student respond?
- What pressures on the student or supervisor might encourage such behaviour?

Conclusions

- Fabrication, falsification, and plagiarism are rare but have a devastating impact on the field concerned
- Questionable research practices are fairly common and harm everyone involved
- Research integrity protects personal reputation, the research field, and society
- Effective prevention: transparency, sound study planning, and ethical behaviour!